Bayes Filter and Kalman Filter Report

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***Abstract*—An autonomous vehicle operating in high-speed racing environments never possesses direct, ground-truth knowledge of its true kinematic or dynamic state. Proprioceptive motion models accumulate unbounded integration drift due to tire slip and surface disturbances, whereas exteroceptive sensors such as LiDAR, stereo cameras, and GPS are corrupted by stochastic measurement noise and varying latencies. Recursive state estimation bridges these two imperfect modalities by maintaining a probabilistic belief distribution over the vehicle's state space. This paper presents a four-part technical review on probabilistic state estimation for autonomous driving. First, the general recursive Bayes filter is derived from the Markov assumption, the Law of Total Probability, and Bayes' rule, with every constituent probabilistic term formally defined. Second, the three foundational linear-Gaussian assumptions that reduce the intractable continuous integrals of the Bayes filter to the closed-form linear Kalman Filter (KF) are examined, alongside the mathematical failure modes introduced by nonlinearities and multimodal distributions. Third, the complete matrix equations of the Kalman Filter prediction and update phases are analyzed, emphasizing the asymptotic behavior of the Kalman gain across extreme sensor noise regimes. Finally, the architectural integration of Extended and Unscented Kalman Filters (EKF/UKF) within the Formula Manipal Driverless (DV) software pipeline is detailed, demonstrating how asynchronous multi-rate sensor fusion and EKF-SLAM enable robust perception and optimal trajectory control.**

***Keywords*—Autonomous Driving, Bayes Filter, Extended Kalman Filter (EKF), Formula Student Driverless, Gaussian Processes, Kalman Gain, Markov Assumption, Sensor Fusion, Simultaneous Localization and Mapping (SLAM), State Estimation.**

I. INTRODUCTION: UNCERTAINTY IN AUTONOMOUS NAVIGATION

State estimation is the foundational discipline of recovering the internal, hidden state vector $\boldsymbol{x}_t$ of a physical system—such as an autonomous vehicle's global position (x, y), heading orientation (θ), longitudinal and lateral velocities (v_x, v_y), and yaw rate (ω)—from imperfect mathematical models and noisy sensor measurements [1], [2]. In autonomous racing applications such as Formula Student Driverless (FSD), a vehicle never knows its true state directly. Instead, it must continuously infer its state in real time from two fundamentally flawed sources of information:

1) Proprioceptive Motion Models (Dead Reckoning): Mathematical kinematic or dynamic models (such as the kinematic or dynamic bicycle model) predict state evolution using known control commands $\boldsymbol{u}_t$ (commanded longitudinal acceleration a and steering angle δ or yaw rate ω) alongside wheel speed encoders and Inertial Measurement Units (IMUs) [2], [3]. Because mathematical models are idealized simplifications of physical reality, and real tires experience longitudinal slip, lateral cornering compliance, uneven track camber, and aerodynamic disturbances, open-loop integration of motion models accumulates unbounded drift over time [1], [2].

2) Exteroceptive Sensor Observations: Environmental perception sensors—including 3D LiDAR, stereo/monocular cameras, radar, and Real-Time Kinematic (RTK) GPS—provide external measurements $\boldsymbol{z}_t$ of landmarks or global coordinates [1], [2]. However, every physical sensor is corrupted by random thermal and environmental noise, finite spatial resolution, optical distortion, and asynchronous communication latencies [2], [4].

If an autonomous controller relies exclusively on dead reckoning, positional drift rapidly causes the vehicle to depart the track boundaries. Conversely, if the controller reacts directly to raw, unsmoothed sensor measurements, high-frequency sensor noise induces violent actuator oscillations and vehicle instability. Probabilistic robotics resolves this dilemma by abandoning single deterministic guesses in favor of a *belief distribution*—a complete probability density function (PDF) over all possible states [1]. State estimation filters recursively fuse the high-rate predictions of the motion model with the drift-free observations of exteroceptive sensors to yield a statistically optimal posterior state estimate [1], [4].

II. THE BAYES FILTER FRAMEWORK

The Bayes filter is the most general mathematical algorithm for calculating recursive belief distributions over an unknown state $\boldsymbol{x}_t$ conditioned on all historical sensor measurements $\boldsymbol{z}_{1:t} = \{\boldsymbol{z}_1, ..., \boldsymbol{z}_t\}$ and control inputs $\boldsymbol{u}_{1:t} = \{\boldsymbol{u}_1, ..., \boldsymbol{u}_t\}$ [1]. Formally, the posterior belief distribution at discrete timestep t is defined as:

A. The Markov Assumption and Conditional Independence

Evaluating (1) directly from raw historical streams would require storing and reprocessing an ever-growing history of sensor readings since $t = 0$, causing computational complexity to grow without bound. To make state estimation recursive, the Bayes filter relies on the *Markov Assumption* (or complete state assumption) [1]. The Markov assumption postulates that past and future data are conditionally independent if the current state $\boldsymbol{x}_t$ is known [1]. Mathematically, this yields two critical conditional independence relations:

In physical terms, (2) states that knowing the previous state $\boldsymbol{x}_{t-1}$ and the current control action $\boldsymbol{u}_t$ is sufficient to predict the next state $\boldsymbol{x}_t$; no earlier states or measurements provide additional predictive power [1]. Similarly, (3) states that the current state $\boldsymbol{x}_t$ completely determines the likelihood of observing sensor measurement $\boldsymbol{z}_t$ [1].

B. The Recursive Two-Step Bayes Filter Algorithm

By combining the Law of Total Probability (marginalization), the Chain Rule of probability, Bayes' Rule, and the Markov assumption (2)–(3), the Bayes filter updates the belief $\text{bel}(\boldsymbol{x}_{t-1})$ from timestep $t-1$ to $\text{bel}(\boldsymbol{x}_t)$ at timestep t through two alternating phases: Prediction and Correction [1], [3].

1) Step 1 — The Prediction Step (Control / Time Update): Before incorporating the new sensor observation $\boldsymbol{z}_t$ at timestep t, the filter processes the control input $\boldsymbol{u}_t$ to project the previous posterior belief $\text{bel}(\boldsymbol{x}_{t-1})$ forward in time [1]. By applying the Law of Total Probability and marginalizing over all possible previous states $\boldsymbol{x}_{t-1}$ in the continuous state space, the filter computes the predicted belief (or *prior belief*) $\overline{\text{bel}}(\boldsymbol{x}_t)$:

Because every physical state transition injects process uncertainty (e.g., wheel slip or actuator imprecision), the convolution integral in (4) spreads and flattens the probability distribution. Consequently, the prediction step strictly increases state uncertainty [1], [2].

2) Step 2 — The Correction Step (Measurement Update): Once the exteroceptive sensor measurement $\boldsymbol{z}_t$ is acquired, the filter corrects the predicted prior belief $\overline{\text{bel}}(\boldsymbol{x}_t)$ using Bayes' Rule [1], [3]. For each hypothetical state $\boldsymbol{x}_t$, the filter multiplies the prior belief $\overline{\text{bel}}(\boldsymbol{x}_t)$ by the measurement likelihood $p(\boldsymbol{z}_t \mid \boldsymbol{x}_t)$ and scales the result by the normalization constant η:

where the normalizer η is the reciprocal of the marginal measurement probability (total evidence) integrated over the state space [1], [3]:

Because multiplying two probability distributions produces a sharper peak with smaller variance than either constituent distribution alone, the correction step injects new sensory information and strictly decreases state uncertainty [1], [2].

C. Comprehensive Dictionary of Bayes Filter Terms

Table I summarizes every probabilistic component of the general Bayes filter algorithm (4)–(6), detailing its mathematical definition and physical role aboard an autonomous vehicle [1]–[3].

TABLE I
MATHEMATICAL AND PHYSICAL DEFINITIONS OF BAYES FILTER TERMS

Term & Symbol	Mathematical Form	Physical Meaning in Autonomous Driving
Initial Belief $\text{bel}(\boldsymbol{x}_0)$	$p(\boldsymbol{x}_0)$	Initial probability distribution at $t = 0$ prior to any motion or sensing. Models certainty at the starting line (narrow peak) or global ignorance (uniform distribution) [1].
Previous Posterior $\text{bel}(\boldsymbol{x}_{t-1})$	$p(\boldsymbol{x}_{t-1} \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t-1})$	Fused probability distribution over the car's state at step $t-1$ after incorporating all prior controls and sensor readings up to $t-1$. Serves as recursive input to step t [1].
Control Input $\boldsymbol{u}_t$	$\boldsymbol{u}_t \in \mathbb{R}^{\wedge}\{m\}$	Vector of known actuation commands applied between $t-1$ and t, such as commanded longitudinal acceleration a and steering yaw rate ω [2].
Motion Model (State Transition)	$p(\boldsymbol{x}_t \mid \boldsymbol{u}_t, \boldsymbol{x}_{t-1})$	Conditional probability density describing vehicle kinematics/dynamics and process noise (tire slip, road bumps). Quantifies transition probability from $\boldsymbol{x}_{t-1}$ to $\boldsymbol{x}_t$ under $\boldsymbol{u}_t$ [1], [2].
Prior Belief $\overline{\text{bel}}(\boldsymbol{x}_t)$	$p(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t})$	Predicted probability distribution over $\boldsymbol{x}_t$ after executing control $\boldsymbol{u}_t$, but before incorporating measurement $\boldsymbol{z}_t$. Exhibits wider spread (higher uncertainty) than $\text{bel}(\boldsymbol{x}_{t-1})$ [1].
Measurement $\boldsymbol{z}_t$	$\boldsymbol{z}_t \in \mathbb{R}^{\wedge}\{k\}$	Vector of noisy exteroceptive observations at time t, such as range-and-bearing (r, φ) to track cones from LiDAR/cameras or position/velocity from GPS [2].
Measurement Likelihood	$p(\boldsymbol{z}_t \mid \boldsymbol{x}_t)$	Generative forward sensor model. Quantifies the likelihood of observing sensor reading $\boldsymbol{z}_t$ assuming the car is truly at state $\boldsymbol{x}_t$, capturing sensor noise and geometry [1], [3].
Normaliser η	$[\int p(\boldsymbol{z}_t \mid \boldsymbol{x}'_t)\, \overline{\text{bel}}(\boldsymbol{x}'_t)\, d\boldsymbol{x}'_t]^{-1}$	Scaling factor required because the product $p(\boldsymbol{z}_t \mid \boldsymbol{x}_t) \cdot \overline{\text{bel}}(\boldsymbol{x}_t)$ does not integrate to 1. Ensures the updated posterior $\text{bel}(\boldsymbol{x}_t)$ is a valid PDF integrating strictly to 1 [1].
Posterior Belief $\text{bel}(\boldsymbol{x}_t)$	$p(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t}, \boldsymbol{u}_{1:t})$	Final corrected state distribution at timestep t after fusing motion prediction with sensor likelihood. Exhibits reduced uncertainty and feeds into step $t+1$ [1].

III. REDUCTION OF THE BAYES FILTER TO THE KALMAN FILTER

Although the general Bayes filter in (4)–(6) is mathematically exact, it cannot be executed directly on an embedded digital computer for arbitrary continuous state spaces [1]. Storing an arbitrary continuous PDF requires infinite memory, and evaluating the marginalization integral in (4) and normalization integral in (6) over an n-dimensional continuous space $\mathbb{R}^{n}$ lacks a closed-form analytical solution for general functions [1].

Invented by Rudolf Emil Kálmán in 1960 [6], the Kalman Filter (KF) circumvents this intractability by restricting the belief representation to multivariate normal (Gaussian) distributions using the *moments representation* [1]. At any timestep t, the entire continuous belief distribution $\text{bel}(\boldsymbol{x}_t)$ is parameterized compactly by just two finite algebraic structures: the mean vector $\boldsymbol{\mu}_t \in \mathbb{R}^{n}$ (also denoted $\hat{\boldsymbol{x}}_{t|t}$) representing the most likely state, and the symmetric positive-definite covariance matrix $\boldsymbol{\Sigma}_t \in \mathbb{R}^{n\times n}$ (also denoted $\boldsymbol{P}_{t|t}$) representing state uncertainty and cross-correlations [1], [2]:

A. The Three Linear-Gaussian Assumptions

In addition to the Markov assumption of the Bayes filter, the posterior belief $\text{bel}(\boldsymbol{x}_t)$ is guaranteed to remain strictly Gaussian for all points in time $t \geq 0$ if and only if the following three foundational assumptions hold [1]:

1) Linear System Dynamics and Linear Measurement Models: Both the state transition function and the sensor observation function must be strictly linear in the state vector $\boldsymbol{x}$ and control vector $\boldsymbol{u}$ [1], [2]. Specifically, the state transition is governed by the linear stochastic difference equation:

and the sensor measurement is governed by the linear observation equation:

where $\boldsymbol{A}_t \in \mathbb{R}^{n\times n}$ (often denoted $\boldsymbol{F}_t$) is the state transition matrix, $\boldsymbol{B}_t \in \mathbb{R}^{n\times m}$ (often denoted $\boldsymbol{G}_t$) is the control input matrix, and $\boldsymbol{C}_t \in \mathbb{R}^{k\times n}$ (often denoted $\boldsymbol{H}_t$) is the observation matrix mapping the state space into the measurement space [1], [2], [4].

2) Zero-Mean White Gaussian Noise: The additive process noise vector $\boldsymbol{\varepsilon}_t \in \mathbb{R}^{n}$ (modeling unmodeled disturbances in motion) and the measurement noise vector $\boldsymbol{\delta}_t \in \mathbb{R}^{k}$ (modeling sensor inaccuracies) must be independent, zero-mean, white Gaussian random processes [1], [2], [4]:

where $\boldsymbol{Q}_t \in \mathbb{R}^{n\times n}$ is the symmetric positive-semi-definite process noise covariance matrix and $\boldsymbol{R}_t \in \mathbb{R}^{k\times k}$ is the symmetric positive-definite measurement noise covariance matrix. (Note on notation: Standard control texts such as Rajamani [5] and Becker [4] denote process noise covariance by $\boldsymbol{Q}_t$ and measurement noise covariance by $\boldsymbol{R}_t$, whereas Corke [2] uses $\boldsymbol{V}$ and $\boldsymbol{W}$, and Thrun et al. [1] swap the symbols $\boldsymbol{R}_t$ and $\boldsymbol{Q}_t$. Throughout this report, $\boldsymbol{Q}_t$ denotes process noise covariance and $\boldsymbol{R}_t$ denotes measurement noise covariance). Under (8)–(10), the motion model and measurement likelihood take the explicit Gaussian forms:

3) Gaussian Initial Belief: The initial belief distribution $\text{bel}(\boldsymbol{x}_0) = p(\boldsymbol{x}_0)$ at $t = 0$ must be normally distributed with initial mean $\boldsymbol{\mu}_0$ and initial covariance matrix $\boldsymbol{\Sigma}_0$ [1]:

B. Why These Assumptions Make the Mathematics Tractable

The linear-Gaussian assumptions render the Bayes filter analytically tractable because multivariate Gaussian distributions satisfy two powerful mathematical closure properties (invariance under affine transformations and multiplication) [1]–[3]:

1) Closure Under Linear Transformation and Convolution (Prediction Step): When a Gaussian random variable $\boldsymbol{x}_{t-1} \sim \mathcal{N}(\boldsymbol{\mu}_{t-1}, \boldsymbol{\Sigma}_{t-1})$ is multiplied by a linear matrix $\boldsymbol{A}_t$, shifted by $\boldsymbol{B}_t\boldsymbol{u}_t$, and convolved with independent additive Gaussian noise $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}_t)$, the exponent inside the Bayes prediction integral (4) is a sum of quadratic forms in $\boldsymbol{x}_{t-1}$ and $\boldsymbol{x}_t$ [1]. Completing the square in the exponent allows the marginalization integral over $\boldsymbol{x}_{t-1}$ to be solved analytically in closed form [1]. The output is guaranteed to be an exact Gaussian prior $\bar{\text{bel}}(\boldsymbol{x}_t) = \mathcal{N}(\boldsymbol{x}_t; \bar{\boldsymbol{\mu}}_t, \bar{\boldsymbol{\Sigma}}_t)$, completely eliminating numerical integration [1].

2) Closure Under Multiplication (Correction Step): In the Bayes measurement update (5), multiplying the Gaussian prior $\mathcal{N}(\boldsymbol{x}_t; \bar{\boldsymbol{\mu}}_t, \bar{\boldsymbol{\Sigma}}_t)$ by the Gaussian likelihood $\mathcal{N}(\boldsymbol{z}_t; \boldsymbol{C}_t\boldsymbol{x}_t, \boldsymbol{R}_t)$ equates to adding their quadratic exponents $J(\boldsymbol{x}_t)$ [1]:

Because the sum of two quadratic polynomials in $\boldsymbol{x}_t$ is itself a quadratic polynomial, $\exp\{-J(\boldsymbol{x}_t)\}$ is guaranteed to be an unnormalized Gaussian distribution [1]. Differentiating $J(\boldsymbol{x}_t)$ with respect to $\boldsymbol{x}_t$ and setting the first derivative $\partial J/\partial \boldsymbol{x}_t = \boldsymbol{0}$ yields the posterior mean $\boldsymbol{\mu}_t$, while the inverse of the second derivative (the Hessian curvature $\partial^2 J/\partial \boldsymbol{x}_t^2$) directly yields the posterior covariance $\boldsymbol{\Sigma}_t = (\boldsymbol{C}_t^{\mathsf{T}}\,\boldsymbol{R}_t^{-1}\,\boldsymbol{C}_t + \bar{\boldsymbol{\Sigma}}_t^{-1})^{-1}$ [1]. Furthermore, the normalization constant η in (6) is absorbed automatically into the standard Gaussian determinant prefactor, bypassing explicit integration entirely [1]. Consequently, infinite-dimensional calculus is reduced to closed-form matrix algebra requiring only $O(k^{2.8} + n^2)$ operations per timestep [1].

C. What Goes Wrong When the Assumptions Fail

In real-world autonomous driving, the three linear-Gaussian assumptions are frequently violated, leading to distinct mathematical and physical failure modes [1], [2]:

1) Violation of Linearity (Warping of Gaussian Distributions): Vehicle kinematics and exteroceptive sensors are inherently nonlinear: $\boldsymbol{x}_t = \boldsymbol{g}(\boldsymbol{u}_t, \boldsymbol{x}_{t-1}) + \boldsymbol{\varepsilon}_t$ and $\boldsymbol{z}_t = \boldsymbol{h}(\boldsymbol{x}_t) + \boldsymbol{\delta}_t$ [1], [2]. Dead-reckoning equations contain trigonometric terms ($v\cos\theta$, $v\sin\theta$), and LiDAR/camera cone observations involve Euclidean range $r = \sqrt{(\Delta x^2 + \Delta y^2)}$ and bearing $\varphi = \text{atan2}(\Delta y, \Delta x)$ [1], [2]. When a Gaussian random variable is passed through a nonlinear function, the output distribution is no longer Gaussian [1], [2]. Nonlinearity stretches one tail of

the bell curve and compresses the other, producing an asymmetric, skewed distribution whose true mean no longer coincides with its peak (mode) [2]. Applying a linear KF to nonlinear dynamics introduces severe linearization bias, causing the estimated covariance $\boldsymbol{P}_{t|t}$ to become inconsistent (overconfident) until the filter diverges completely [1], [2].

2) Violation of Gaussian Noise (Outliers and Heavy Tails): Physical sensors suffer from non-Gaussian errors: LiDAR beams reflect off wet asphalt puddles, stereo vision algorithms produce false-positive cone detections due to sun glare, and wheel encoders experience asymmetric spikes during wheel lockup [1], [2]. Because a Gaussian distribution has exponentially decaying tails, quadratic least-squares estimators like the KF are extremely sensitive to outliers; a single unrejected outlier measurement pulls the posterior mean violently off course [1], [2].

3) Violation of Unimodal Belief (Multimodal Hypotheses & Correspondence Errors): A Gaussian distribution is strictly unimodal—it possesses a single peak annotated by a single uncertainty ellipse [1]. It cannot represent simultaneous, distinct hypotheses (e.g., global localization uncertainty when a car observes identical cone geometries at two different hairpins, or ambiguous data association where the car sees cone *i* but mistakes it for cone *j*) [1], [2].

4) Engineering Remedies: To handle mild-to-moderate nonlinearities, the Extended Kalman Filter (EKF) linearizes $\boldsymbol{g}(\cdot)$ and $\boldsymbol{h}(\cdot)$ around the current mean estimate using first-order Taylor series expansions (Jacobian matrices $\boldsymbol{G}_t = \partial \boldsymbol{g}/\partial \boldsymbol{x}$ and $\boldsymbol{H}_t = \partial \boldsymbol{h}/\partial \boldsymbol{x}$) [1], [2]. For stronger nonlinearities, the Unscented Kalman Filter (UKF) propagates deterministic sample points ('sigma points') through the exact nonlinear functions to capture mean and covariance to second/third order without Jacobians [2]. For multimodal beliefs or non-Gaussian noise, nonparametric Particle Filters (Monte Carlo Localization) are employed [1], [2].

IV. THE KALMAN FILTER EQUATIONS AND ASYMPTOTIC GAIN ANALYSIS

Using standard state-space control notation where $\hat{\boldsymbol{x}}_{t|t-1}$ (or $\bar{\boldsymbol{\mu}}_t$) and $\boldsymbol{P}_{t|t-1}$ (or $\bar{\boldsymbol{\Sigma}}_t$) denote the prior mean and covariance at step t given measurements up to $t-1$, and $\hat{\boldsymbol{x}}_{t|t}$ (or $\boldsymbol{\mu}_t$) and $\boldsymbol{P}_{t|t}$ (or $\boldsymbol{\Sigma}_t$) denote the posterior mean and covariance given measurements up to t, the linear Kalman Filter executes in six sequential matrix equations grouped into two phases [1], [2], [4].

A. Phase 1: The Prediction Step (Time / Control Update)

The prediction step is an open-loop propagation of the state belief across timestep Δt driven solely by the system dynamics model ($\boldsymbol{A}_t$, $\boldsymbol{B}_t$) and the known control vector $\boldsymbol{u}_t$ [2], [4]:

1) Role of the State Prediction Equation (15): Equation (15) projects the previous posterior state estimate $\hat{\boldsymbol{x}}_{t-1|t-1}$ forward in time using the deterministic state transition matrix $\boldsymbol{A}_t$ and control coupling matrix $\boldsymbol{B}_t$ [1], [2]. Because the process noise $\boldsymbol{\varepsilon}_t$ has zero mean ($\mathrm{E}[\boldsymbol{\varepsilon}_t] = \boldsymbol{0}$), it does not appear in the mean propagation equation [1].

2) Role of the Covariance Prediction Equation (16): Equation (16) predicts how the state uncertainty ellipse evolves during motion [1], [2]. Because covariance is quadratic with respect to linear transformations—specifically, $\mathrm{Var}(\boldsymbol{A}_t\boldsymbol{x}) = \boldsymbol{A}_t\,\mathrm{Var}(\boldsymbol{x})\,\boldsymbol{A}_t^{\mathrm{T}}$—the transition matrix $\boldsymbol{A}_t$ multiplies $\boldsymbol{P}_{t-1|t-1}$ on the left and its transpose $\boldsymbol{A}_t^{\mathrm{T}}$ multiplies on the right, scaling and rotating the uncertainty ellipse according to system dynamics [1], [2]. Crucially, the process noise covariance matrix $\boldsymbol{Q}_t$ is added to account for unmodeled disturbances [2], [4]. Because both $\boldsymbol{A}_t\boldsymbol{P}_{t-1|t-1}\boldsymbol{A}_t^{\mathrm{T}}$ and $\boldsymbol{Q}_t$ are positive-definite matrices, their sum guarantees that the predicted uncertainty $\boldsymbol{P}_{t|t-1}$ increases during dead reckoning in the absence of sensor measurements [1], [2].

B. Phase 2: The Update Step (Measurement Correction)

When a sensor observation $\boldsymbol{z}_t$ arrives, the update step fuses $\boldsymbol{z}_t$ with the prior prediction to compute the posterior state and shrink the uncertainty [1], [2], [4]:

1) Role of the Innovation and Innovation Covariance (17): In (17), the term $\hat{\boldsymbol{z}}_t = \boldsymbol{C}_t\hat{\boldsymbol{x}}_{t|t-1}$ projects our predicted state into the sensor's measurement space to compute what sensor reading we expect to observe [1], [2]. The difference $\boldsymbol{v}_t = \boldsymbol{z}_t - \boldsymbol{C}_t\hat{\boldsymbol{x}}_{t|t-1}$ is termed the *innovation* (or measurement residual) because it represents the new information introduced by the sensor that was not anticipated by the motion model [2], [4]. The matrix $\boldsymbol{S}_t \in \mathbb{R}^{\{k\times k\}}$ is the *innovation covariance*; the term $\boldsymbol{C}_t\boldsymbol{P}_{t|t-1}\boldsymbol{C}_t^{\mathrm{T}}$ projects the prior state uncertainty into measurement space, and adding the sensor noise covariance $\boldsymbol{R}_t$ yields the total expected uncertainty of the residual $\boldsymbol{v}_t$ [2].

2) Role of the Kalman Gain Equation (18): The matrix $\boldsymbol{K}_t \in \mathbb{R}^{\{n\times k\}}$ is the optimal *Kalman Gain* [1], [4]. Derived by minimizing the trace of the posterior error covariance matrix $\mathrm{Tr}(\boldsymbol{P}_{t|t})$ (the Minimum Mean-Square Error criterion), $\boldsymbol{K}_t$ acts as a statistical weighting factor that determines how heavily the measurement residual $\boldsymbol{v}_t$ should be trusted relative to the prior prediction $\hat{\boldsymbol{x}}_{t|t-1}$ [2], [4]. Structurally, $\boldsymbol{K}_t$ is the ratio of the cross-covariance between state and measurement ($\boldsymbol{P}_{t|t-1}\boldsymbol{C}_t^{\mathrm{T}}$) to the total measurement innovation covariance ($\boldsymbol{S}_t$) [2].

3) Role of the State Estimate Update Equation (19): Equation (19) computes the corrected posterior state $\hat{\boldsymbol{x}}_{t|t}$ by taking the prior prediction $\hat{\boldsymbol{x}}_{t|t-1}$ and adding a correction term $\boldsymbol{K}_t\boldsymbol{v}_t$ proportional to both the Kalman gain and the innovation [1], [2], [4]. By rearranging (19) into the equivalent weighted-average form $\hat{\boldsymbol{x}}_{t|t} = (\boldsymbol{I} - \boldsymbol{K}_t\boldsymbol{C}_t)\hat{\boldsymbol{x}}_{t|t-1} + \boldsymbol{K}_t\boldsymbol{z}_t$, it becomes evident that the posterior estimate is a convex linear combination of the prediction and the measurement [4].

4) Role of the Error Covariance Update Equation (20): Equation (20) updates the state uncertainty to reflect the information gain from the sensor measurement [1], [2]. Because $\boldsymbol{K}_t\boldsymbol{C}_t\boldsymbol{P}_{t|t-1}$ is subtracted from the prior covariance $\boldsymbol{P}_{t|t-1}$, the uncertainty ellipse is strictly deflated (shrunk) across all observed state dimensions [1], [2]. In practical floating-point

implementations, subtracting two matrices in (20) can occasionally cause $\boldsymbol{P}_{t|t}$ to lose symmetry or positive-definiteness due to numerical roundoff; therefore, automotive and aerospace code often implements the algebraically equivalent *Joseph Form* [2], [4]:

which is a sum of two symmetric positive-definite quadratic forms and guarantees numerical stability [2], [4].

C. Behavior of the Kalman Gain: Very Noisy vs. Very Accurate Sensors

To understand how the Kalman Gain $\boldsymbol{K}_t$ dynamically arbitrates between model prediction and sensor observation, consider the mathematical limits of (18)–(20) as the sensor measurement noise covariance $\boldsymbol{R}_t$ varies between extreme regimes (summarized in Table II) [2], [4]:

1) Case 1 — The Sensor is Very Noisy (R_t → ∞, or Prior Uncertainty P_{t|t−1} → 0): When a sensor is severely corrupted by noise, the diagonal entries (variances) of $\boldsymbol{R}_t$ become very large ($\boldsymbol{R}_t \to \infty$) relative to the projected state uncertainty $\boldsymbol{C}_t\boldsymbol{P}_{t|t-1}\boldsymbol{C}_t^T$ [4]. In the Kalman gain equation (18), the inverted matrix $\boldsymbol{S}_t^{-1} = (\boldsymbol{C}_t\boldsymbol{P}_{t|t-1}\boldsymbol{C}_t^T + \boldsymbol{R}_t)^{-1}$ approaches the zero matrix $\boldsymbol{0}$. Consequently:

Substituting $\boldsymbol{K}_t = \boldsymbol{0}$ into the state update (19) and covariance update (20) yields $\hat{\boldsymbol{x}}_{t|t} = \hat{\boldsymbol{x}}_{t|t-1} + \boldsymbol{0} \cdot \boldsymbol{v}_t = \hat{\boldsymbol{x}}_{t|t-1}$ and $\boldsymbol{P}_{t|t} = (\boldsymbol{I} - \boldsymbol{0})\boldsymbol{P}_{t|t-1} = \boldsymbol{P}_{t|t-1}$. Physically, the filter completely ignores the noisy sensor reading $\boldsymbol{z}_t$ and trusts its internal kinematic motion model 100%, leaving the prior uncertainty unchanged [4].

2) Case 2 — The Sensor is Very Accurate (R_t → 0, or Prior Uncertainty P_{t|t−1} → ∞): Conversely, when a sensor is virtually noise-free ($\boldsymbol{R}_t \to \boldsymbol{0}$) or the vehicle's prior position uncertainty is huge ($\boldsymbol{P}_{t|t-1} \to \infty$), the noise term $\boldsymbol{R}_t$ vanishes from the denominator of (18). Assuming $\boldsymbol{C}_t$ is invertible, the Kalman gain reduces to the inverse of the observation matrix [4]:

Substituting $\boldsymbol{K}_t = \boldsymbol{C}_t^{-1}$ into the state update (19) yields:

and the posterior covariance (20) collapses to $\boldsymbol{P}_{t|t} = (\boldsymbol{I} - \boldsymbol{C}_t^{-1}\boldsymbol{C}_t)\boldsymbol{P}_{t|t-1} = \boldsymbol{0}$. Physically, the prior kinematic prediction $\hat{\boldsymbol{x}}_{t|t-1}$ cancels out completely; the filter discards the motion model prediction and snaps its posterior state estimate directly to the inverted sensor measurement $\boldsymbol{C}_t^{-1}\boldsymbol{z}_t$ with near-zero uncertainty [4].

TABLE II
ASYMPTOTIC BEHAVIOR OF THE KALMAN GAIN ACROSS SENSOR NOISE REGIMES

Operating Regime	Noise / Covariance Ratio	Kalman Gain Limit ($\boldsymbol{K}_t$)	Posterior State & Covariance	Filter Physical Action
Very Noisy Sensor (or Perfect Model)	$\boldsymbol{R}_t \gg \boldsymbol{C}_t\boldsymbol{P}_{t\|t-1}\boldsymbol{C}_t^T$ ($\boldsymbol{R}_t \to \infty$)	$\boldsymbol{K}_t \to \boldsymbol{0}$	$\hat{\boldsymbol{x}}_{t\|t} \approx \hat{\boldsymbol{x}}_{t\|t-1}$ $\boldsymbol{P}_{t\|t} \approx \boldsymbol{P}_{t\|t-1}$	Rejects corrupted sensor measurement $\boldsymbol{z}_t$; relies entirely on kinematic dead reckoning [4].
Balanced Uncertainty (Nominal Operation)	$\boldsymbol{R}_t \approx \boldsymbol{C}_t\boldsymbol{P}_{t\|t-1}\boldsymbol{C}_t^T$ (Scalar: $K_t \approx 0.5$)	$\boldsymbol{0} < \boldsymbol{K}_t < \boldsymbol{C}_t^{-1}$	$\hat{\boldsymbol{x}}_{t\|t} = \hat{\boldsymbol{x}}_{t\|t-1} + \boldsymbol{K}_t\boldsymbol{v}_t$ $\boldsymbol{P}_{t\|t} < \boldsymbol{P}_{t\|t-1}$	Optimal MMSE fusion of kinematic model and sensor; posterior variance is smaller than both [1], [4].
Very Accurate Sensor (or High Prior Drift)	$\boldsymbol{R}_t \ll \boldsymbol{C}_t\boldsymbol{P}_{t\|t-1}\boldsymbol{C}_t^T$ ($\boldsymbol{R}_t \to \boldsymbol{0}$)	$\boldsymbol{K}_t \to \boldsymbol{C}_t^{-1}$ (Scalar: $K_t \to 1$)	$\hat{\boldsymbol{x}}_{t\|t} \approx \boldsymbol{C}_t^{-1}\boldsymbol{z}_t$ $\boldsymbol{P}_{t\|t} \to \boldsymbol{0}$	Overrides drifted kinematic prediction; snaps state directly to sensor observation $\boldsymbol{z}_t$ [4].

V. RELEVANCE TO THE FORMULA MANIPAL DV PIPELINE

Within the Formula Manipal Driverless Vehicle (DV) autonomous software architecture, Kalman filtering (implemented as an Extended or Unscented Kalman Filter due to nonlinear vehicle kinematics) serves as the central hub connecting raw hardware sensors and perception modules to Path Planning and Optimal Control [1], [2].

A. Nonlinear EKF Generalization for Vehicle Kinematics

Because an autonomous racecar moves along circular and clothoid arcs governed by trigonometric functions, the linear equations (15)–(20) are generalized to the Extended Kalman Filter (EKF) [1], [2]:

where $\boldsymbol{G}_t = \partial\boldsymbol{g}/\partial\boldsymbol{x}|_{_}\{\hat{\boldsymbol{x}}_{t-1|t-1}\}$ and $\boldsymbol{H}_t = \partial\boldsymbol{h}/\partial\boldsymbol{x}|_{_}\{\hat{\boldsymbol{x}}_{t|t-1}\}$ are the state transition and measurement Jacobian matrices evaluated at the current state estimate [1], [2]. As summarized in Table III, this filter architecture operates across three distinct layers of the Formula Manipal DV pipeline.

B. Asynchronous Multi-Rate Ego-State & Bias Estimation

To execute high-speed trajectory tracking, the DV controller requires continuous, low-latency estimates of the ego-vehicle state vector, including longitudinal/lateral velocities and IMU sensor biases [5]:

1) High-Rate Proprioceptive Prediction (100–400 Hz): The onboard IMU and wheel speed encoders stream accelerations, yaw rates, and wheel angular velocities at 100–400 Hz. However, raw IMU accelerations suffer from mechanical vibration noise and temperature-induced bias drift (b_{ax}, b_{ay}) [5], while wheel speed sensors overestimate longitudinal velocity v_x during wheel spin (launch acceleration) or lockup (heavy braking). The EKF runs its Prediction Step (25) at the full 100–400 Hz IMU rate using a kinematic or dynamic

bicycle model, supplying smooth, low-latency state predictions to the controller every $\Delta t = 0.005$–0.02 s [2].

2) Asynchronous Exteroceptive Updates (10–50 Hz): Because the Kalman filter is inherently asynchronous [2], whenever a slower measurement arrives from RTK-GPS, an optical ground-speed sensor, or visual/LiDAR odometry, the filter triggers the Update Step (26)–(27) using that sensor's specific Jacobian $\boldsymbol{H}_t$ and noise covariance $\boldsymbol{R}_t$ [2], [5]. Following Rajamani's state-augmentation formulation [5], the cross-correlations in $\boldsymbol{P}_{t|t}$ allow the filter to observe and subtract out the accelerometer and gyroscope biases in real time.

C. Local Cone Perception Tracking and EKF-SLAM

In Formula Student Driverless events, track boundaries are marked by Blue cones on the left and Yellow cones on the right. The LiDAR clustering and stereo camera neural-network pipelines detect these cones in the local vehicle reference frame:

1) Local Multi-Object Cone Tracking: Before landmarks are committed to the global map, individual linear Kalman Filters track each detected cone in the local car frame across consecutive camera/LiDAR frames (10–20 Hz). This smooths out bounding-box jitter and rejects transient false-positive 'ghost' detections using Mahalanobis distance gating on the innovation covariance $\boldsymbol{S}_t$: $\boldsymbol{v}_t^{\mathrm{T}}\, \boldsymbol{S}_t^{-1}\, \boldsymbol{v}_t \le \chi^2_{\mathrm{gate}}$ [1], [2].

2) EKF-SLAM and Global Localization: To localize the vehicle globally and build a map of an unknown track during Autocross or Trackdrive, the EKF state vector is augmented to contain both the 3-DOF vehicle pose $\boldsymbol{x}_c = [x, y, \theta]^{\mathrm{T}}$ and the global coordinates of M observed track cones $\boldsymbol{m} = [x_1, y_1, \ldots, x_M, y_M]^{\mathrm{T}}$, yielding a state vector of dimension $3 + 2M$ [1], [2]. In the update step, each expected cone position $[x_i, y_i]^{\mathrm{T}}$ is transformed from the global frame into the car's local frame via the 2D rotation matrix $\boldsymbol{R}(-\theta_t)$ (the exact frame transformation from Task 6 Q1):

When the car re-observes known cones (loop closure), the innovation $\boldsymbol{v}_t$ between the actual LiDAR/camera cone detection and the predicted local cone position (29) updates the entire augmented state vector via the off-diagonal cross-correlations of $\boldsymbol{P}_{t|t}$, simultaneously correcting vehicle odometry drift and refining the global track map [1], [2].

D. Upstream Integration with the Optimal Controller (MPC)

Finally, the Kalman filter sits immediately upstream of Path Planning and the Model Predictive Controller (MPC) formulated in Task 6 Q2, which minimizes the multi-step quadratic cost function over horizon N:

Evaluating the heading error $e_{h,k}$, the cross-track error $e_{ct,k}$ relative to the waypoint line, and the velocity error $e_{v,k}$ requires an accurate, noise-free initial state $[x_0, y_0, \theta_0, v_0]^{\mathrm{T}}$ at the start of every control cycle. Without the Kalman filter, pure dead-reckoning drift causes the car to steer off-track while believing it is on the centerline, whereas feeding raw noisy sensor measurements directly into (30) causes the optimizer to command violent, chattering steering (ω^*) and acceleration (a^*) spikes that saturate the actuators. By fusing high-rate kinematic predictions with drift-free exteroceptive measurements, the Kalman Filter ensures smooth, optimal autonomous control.

TABLE III
KALMAN FILTER INTEGRATION ACROSS THE FORMULA MANIPAL DV SOFTWARE PIPELINE

DV Pipeline Subsystem	Filter Architecture	Prediction Inputs ($\boldsymbol{u}_t$)	Measurement Updates ($\boldsymbol{z}_t$)	Output to Downstream Modules
1. Ego-Velocity & State Estimation	Asynchronous Error-State EKF / UKF (100–400 Hz)	IMU accelerations (a_x, a_y), gyro yaw rate (ω), steering angle (δ)	Wheel encoders, RTK-GPS, Optical speed sensor, Visual/LiDAR odometry	Clean ego-velocity (v_x, v_y), yaw rate ω, and bias-corrected accelerations for MPC (Q2).
2. Local Perception Cone Tracking	Linear Kalman Filter + Mahalanobis Gate (10–20 Hz)	Constant-position (in global) / ego-motion compensated local model	3D LiDAR cluster centroids & YOLO stereo camera cone bounding boxes	Filtered, jitter-free local cone coordinates (x_{local}, y_{local}) with ghost/outlier rejection.
3. Localization & Mapping (SLAM)	EKF-SLAM / FastSLAM / Graph-SLAM (10–20 Hz)	Fused vehicle odometry (v_x, v_y, ω) propagating 3-DOF car pose	Matched Blue & Yellow cone range/bearing observations via $\boldsymbol{R}(-\theta)$ (Q1)	Drift-free global car pose (x, y, θ) and global track map for Delaunay path planning.

FORMULAS

$$\mathrm{bel}(\boldsymbol{x}_t) = p(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t}, \boldsymbol{u}_{1:t}) \tag{1}$$

$$p(\boldsymbol{x}_t \mid \boldsymbol{x}_{0:t-1}, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) = p(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}, \boldsymbol{u}_t) \tag{2}$$

$$p(\boldsymbol{z}_t \mid \boldsymbol{x}_{0:t}, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) = p(\boldsymbol{z}_t \mid \boldsymbol{x}_t) \tag{3}$$

$$\overline{\mathrm{bel}}(\boldsymbol{x}_t) = p(\boldsymbol{x}_t \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) = \int p(\boldsymbol{x}_t \mid \boldsymbol{u}_t, \boldsymbol{x}_{t-1})\, \mathrm{bel}(\boldsymbol{x}_{t-1})\, d\boldsymbol{x}_{t-1} \tag{4}$$

$$\mathrm{bel}(\boldsymbol{x}_t) = \eta \cdot p(\boldsymbol{z}_t \mid \boldsymbol{x}_t) \cdot \overline{\mathrm{bel}}(\boldsymbol{x}_t) \tag{5}$$

$$\eta = [\, p(\boldsymbol{z}_t \mid \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t})\,]^{-1} = [\, \int p(\boldsymbol{z}_t \mid \boldsymbol{x}'_t)\, \overline{\mathrm{bel}}(\boldsymbol{x}'_t)\, d\boldsymbol{x}'_t\,]^{-1} \tag{6}$$

$$\text{bel}(\boldsymbol{x}_t) = \mathcal{N}(\boldsymbol{x}_t;\, \boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t) = \det(2\pi\boldsymbol{\Sigma}_t)^{-1/2} \exp\{ -1/2\ (\boldsymbol{x}_t - \boldsymbol{\mu}_t)^T\ \boldsymbol{\Sigma}_t^{-1}\ (\boldsymbol{x}_t - \boldsymbol{\mu}_t)\ \} \tag{7}$$

$$\boldsymbol{x}_t = \boldsymbol{A}_t\, \boldsymbol{x}_{t-1} + \boldsymbol{B}_t\, \boldsymbol{u}_t + \boldsymbol{\varepsilon}_t \tag{8}$$

$$\boldsymbol{z}_t = \boldsymbol{C}_t\, \boldsymbol{x}_t + \boldsymbol{\delta}_t \tag{9}$$

$$\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}_t) \quad \text{and} \quad \boldsymbol{\delta}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_t) \tag{10}$$

$$p(\boldsymbol{x}_t \mid \boldsymbol{u}_t, \boldsymbol{x}_{t-1}) = \det(2\pi\boldsymbol{Q}_t)^{-1/2} \exp\{ -1/2\ (\boldsymbol{x}_t - \boldsymbol{A}_t\boldsymbol{x}_{t-1} - \boldsymbol{B}_t\boldsymbol{u}_t)^T\ \boldsymbol{Q}_t^{-1}\ (\boldsymbol{x}_t - \boldsymbol{A}_t\boldsymbol{x}_{t-1} - \boldsymbol{B}_t\boldsymbol{u}_t)\ \} \tag{11}$$

$$p(\boldsymbol{z}_t \mid \boldsymbol{x}_t) = \det(2\pi\boldsymbol{R}_t)^{-1/2} \exp\{ -1/2\ (\boldsymbol{z}_t - \boldsymbol{C}_t\boldsymbol{x}_t)^T\ \boldsymbol{R}_t^{-1}\ (\boldsymbol{z}_t - \boldsymbol{C}_t\boldsymbol{x}_t)\ \} \tag{12}$$

$$\text{bel}(\boldsymbol{x}_0) = \mathcal{N}(\boldsymbol{x}_0;\, \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) = \det(2\pi\boldsymbol{\Sigma}_0)^{-1/2} \exp\{ -1/2\ (\boldsymbol{x}_0 - \boldsymbol{\mu}_0)^T\ \boldsymbol{\Sigma}_0^{-1}\ (\boldsymbol{x}_0 - \boldsymbol{\mu}_0)\ \} \tag{13}$$

$$J(\boldsymbol{x}_t) = 1/2\ (\boldsymbol{z}_t - \boldsymbol{C}_t\boldsymbol{x}_t)^T\ \boldsymbol{R}_t^{-1}\ (\boldsymbol{z}_t - \boldsymbol{C}_t\boldsymbol{x}_t) + 1/2\ (\boldsymbol{x}_t - \bar{\boldsymbol{\mu}_t})^T\ \bar{\boldsymbol{\Sigma}_t}^{-1}\ (\boldsymbol{x}_t - \bar{\boldsymbol{\mu}_t}) \tag{14}$$

$$\hat{\boldsymbol{x}_{t|t-1}} = \boldsymbol{A}_t\, \hat{\boldsymbol{x}_{t-1|t-1}} + \boldsymbol{B}_t\, \boldsymbol{u}_t \tag{15}$$

$$\boldsymbol{P}_{t|t-1} = \boldsymbol{A}_t\, \boldsymbol{P}_{t-1|t-1}\, \boldsymbol{A}_t^T + \boldsymbol{Q}_t \tag{16}$$

$$\boldsymbol{v}_t = \boldsymbol{z}_t - \boldsymbol{C}_t\, \hat{\boldsymbol{x}_{t|t-1}} \quad \text{and} \quad \boldsymbol{S}_t = \boldsymbol{C}_t\, \boldsymbol{P}_{t|t-1}\, \boldsymbol{C}_t^T + \boldsymbol{R}_t \tag{17}$$

$$\boldsymbol{K}_t = \boldsymbol{P}_{t|t-1}\, \boldsymbol{C}_t^T\, \boldsymbol{S}_t^{-1} = \boldsymbol{P}_{t|t-1}\, \boldsymbol{C}_t^T\, (\boldsymbol{C}_t\, \boldsymbol{P}_{t|t-1}\, \boldsymbol{C}_t^T + \boldsymbol{R}_t)^{-1} \tag{18}$$

$$\hat{\boldsymbol{x}_{t|t}} = \hat{\boldsymbol{x}_{t|t-1}} + \boldsymbol{K}_t\, (\boldsymbol{z}_t - \boldsymbol{C}_t\, \hat{\boldsymbol{x}_{t|t-1}}) = \hat{\boldsymbol{x}_{t|t-1}} + \boldsymbol{K}_t\, \boldsymbol{v}_t \tag{19}$$

$$\boldsymbol{P}_{t|t} = (\boldsymbol{I} - \boldsymbol{K}_t\, \boldsymbol{C}_t)\, \boldsymbol{P}_{t|t-1} \tag{20}$$

$$\boldsymbol{P}_{t|t} = (\boldsymbol{I} - \boldsymbol{K}_t\, \boldsymbol{C}_t)\, \boldsymbol{P}_{t|t-1}\, (\boldsymbol{I} - \boldsymbol{K}_t\, \boldsymbol{C}_t)^T + \boldsymbol{K}_t\, \boldsymbol{R}_t\, \boldsymbol{K}_t^T \tag{21}$$

$$\lim_\{\boldsymbol{R}_t \to \infty\}\ \boldsymbol{K}_t = \boldsymbol{P}_{t|t-1}\, \boldsymbol{C}_t^T\, (\boldsymbol{C}_t\, \boldsymbol{P}_{t|t-1}\, \boldsymbol{C}_t^T + \boldsymbol{R}_t)^{-1} = \boldsymbol{0} \tag{22}$$

$$\lim_\{\boldsymbol{R}_t \to \boldsymbol{0}\}\ \boldsymbol{K}_t = \boldsymbol{P}_{t|t-1}\, \boldsymbol{C}_t^T\, (\boldsymbol{C}_t\, \boldsymbol{P}_{t|t-1}\, \boldsymbol{C}_t^T)^{-1} = \boldsymbol{C}_t^{-1} \tag{23}$$

$$\hat{\boldsymbol{x}_{t|t}} = \hat{\boldsymbol{x}_{t|t-1}} + \boldsymbol{C}_t^{-1}\, (\boldsymbol{z}_t - \boldsymbol{C}_t\, \hat{\boldsymbol{x}_{t|t-1}}) = \hat{\boldsymbol{x}_{t|t-1}} + \boldsymbol{C}_t^{-1}\, \boldsymbol{z}_t - \hat{\boldsymbol{x}_{t|t-1}} = \boldsymbol{C}_t^{-1}\, \boldsymbol{z}_t \tag{24}$$

$$\hat{\boldsymbol{x}_{t|t-1}} = \boldsymbol{g}(\boldsymbol{u}_t, \hat{\boldsymbol{x}_{t-1|t-1}}) \quad \text{and} \quad \boldsymbol{P}_{t|t-1} = \boldsymbol{G}_t\, \boldsymbol{P}_{t-1|t-1}\, \boldsymbol{G}_t^T + \boldsymbol{Q}_t \tag{25}$$

$$\boldsymbol{K}_t = \boldsymbol{P}_{t|t-1}\, \boldsymbol{H}_t^T\, (\boldsymbol{H}_t\, \boldsymbol{P}_{t|t-1}\, \boldsymbol{H}_t^T + \boldsymbol{R}_t)^{-1} \tag{26}$$

$$\hat{\boldsymbol{x}_{t|t}} = \hat{\boldsymbol{x}_{t|t-1}} + \boldsymbol{K}_t\, (\boldsymbol{z}_t - \boldsymbol{h}(\hat{\boldsymbol{x}_{t|t-1}})) \quad \text{and} \quad \boldsymbol{P}_{t|t} = (\boldsymbol{I} - \boldsymbol{K}_t\, \boldsymbol{H}_t)\, \boldsymbol{P}_{t|t-1} \tag{27}$$

$$\boldsymbol{x}_{\text{ego}} = [\ x,\ y,\ \theta,\ v_x,\ v_y,\ \omega,\ b_{ax},\ b_{ay},\ b_{\omega}\ ]^T \tag{28}$$

$$\boldsymbol{h}_i(\boldsymbol{x}_t) = \boldsymbol{R}(-\theta_t)\ (\ [x_i, y_i]^T - [x_t, y_t]^T\ ) \tag{29}$$

$$J(a, \omega) = \sum_\{k=1\}\^\{N\}\ [\ W_h\ e_{h,k}^2 + W_{ct}\ e_{ct,k}^2 + W_v\ (v_k - v_{des})^2\ ] \tag{30}$$

VI. CONCLUSION

Probabilistic state estimation forms the mathematical backbone of autonomous navigation. Starting from the general recursive Bayes filter—which propagates arbitrary belief distributions via the motion model in the prediction step and refines them via the measurement likelihood and normalizer in the correction step—this report demonstrated how imposing linear models, zero-mean Gaussian noise, and a Gaussian initial belief reduces intractable continuous integrals to the closed-form matrix operations of the Kalman Filter. By dynamically adjusting the Kalman gain $\boldsymbol{K}_t$ based on the ratio of state uncertainty to sensor noise covariance, the filter seamlessly shifts trust between kinematic dead reckoning and exteroceptive observations. Within the Formula Manipal DV pipeline, extending this framework via EKF/UKF state estimation and SLAM bridges noisy perception hardware with optimal trajectory control, enabling reliable, high-speed autonomous racing.

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